21l-5692

April 7, 2024

1 Muhammad Hassan Khalid

2 21L-5692

3 Question 1

[53]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import urduhack
from urduhack.normalization import normalize
from urduhack.preprocessing import remove_punctuation, remove_accents,␣
↪replace_emails, replace_numbers, replace_currency_symbols
from urduhack.models.lemmatizer import lemmatizer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import precision_recall_fscore_support
from sklearn.metrics import classification_report, confusion_matrix
from keras.preprocessing.sequence import pad_sequences
from keras.preprocessing.text import Tokenizer
from keras.models import Sequential
from keras.layers import Embedding, LSTM, GRU, Bidirectional, Dense, Dropout

# Download Urdu resources
urduhack.download()

# Read the dataset
urdu_dataset = pd.read_csv("urdu-sentiment-corpus-v1.tsv", sep="\t")

# Remove null values
urdu_dataset.dropna(inplace=True)

# Label Encoding
label_encoder = LabelEncoder()
urdu_dataset['Encoded_Class'] = label_encoder.
↪fit_transform(urdu_dataset['Class'])
```

```python
# Preprocessing
def preprocess_text(text):
    text = normalize(text)
    text = remove_accents(text)
    text = remove_punctuation(text)
    text = replace_emails(text)
    text = replace_numbers(text)
    text = replace_currency_symbols(text)
    return text

urdu_dataset['cleaned_text'] = urdu_dataset['Tweet'].apply(preprocess_text)

# Tokenization
tokenizer = Tokenizer()
tokenizer.fit_on_texts(urdu_dataset['cleaned_text'])
sequences = tokenizer.texts_to_sequences(urdu_dataset['cleaned_text'])
max_sequence_length = max(len(x) for x in sequences)
padded_sequences = pad_sequences(sequences, maxlen=max_sequence_length,
 ↪padding='post')

# Splitting the dataset
X_train, X_test, y_train, y_test = train_test_split(padded_sequences,
 ↪urdu_dataset['Encoded_Class'], test_size=0.25, random_state=42)

# Model building and evaluation
def build_model(model_type, num_layers, dropout_rate):
    model = Sequential()
    model.add(Embedding(input_dim=len(tokenizer.word_index) + 1,
 ↪output_dim=128, input_length=max_sequence_length))

    for _ in range(num_layers):
        if model_type == 'RNN':
            model.add(LSTM(units=64, return_sequences=True))
        elif model_type == 'GRU':
            model.add(GRU(units=64, return_sequences=True))
        elif model_type == 'LSTM':
            model.add(LSTM(units=64, return_sequences=True))
        elif model_type == 'BiLSTM':
            model.add(Bidirectional(LSTM(units=64, return_sequences=True)))
        else:
            raise ValueError(f"Invalid model_type: {model_type}")
        model.add(Dropout(dropout_rate))

    if model_type == 'BiLSTM':
        model.add(Bidirectional(LSTM(units=64)))
    else:
```

```python
        model.add(LSTM(units=64))
    model.add(Dropout(dropout_rate))
    model.add(Dense(1, activation='sigmoid'))

    model.compile(optimizer='adam', loss='binary_crossentropy',␣
↪metrics=['accuracy'])
    return model

def evaluate_model(model, X_test, y_test):
    y_pred = (model.predict(X_test) > 0.5).astype("int32")
    print(classification_report(y_test, y_pred))
    cm = confusion_matrix(y_test, y_pred)
    sns.heatmap(cm, annot=True, fmt='d')
    plt.xlabel('Predicted Label')
    plt.ylabel('Actual Label')
    plt.title('Confusion Matrix')
    plt.show()

# Model evaluation with different parameters
parameters = [('RNN', 2, 0.3), ('RNN', 2, 0.7), ('GRU', 2, 0.3), ('GRU', 2, 0.
↪7),
              ('LSTM', 2, 0.3), ('LSTM', 2, 0.7), ('BiLSTM', 2, 0.3),␣
↪('BiLSTM', 2, 0.7),
              ('RNN', 3, 0.3), ('RNN', 3, 0.7), ('GRU', 3, 0.3), ('GRU', 3, 0.
↪7),
              ('LSTM', 3, 0.3), ('LSTM', 3, 0.7), ('BiLSTM', 3, 0.3),␣
↪('BiLSTM', 3, 0.7)]

results = []

for param in parameters:
    model_type, num_layers, dropout_rate = param
    model = build_model(model_type, num_layers, dropout_rate)
    history = model.fit(X_train, y_train, epochs=8, batch_size=32,␣
↪validation_split=0.25, verbose=0)
    loss, accuracy = model.evaluate(X_test, y_test, verbose=0)
    y_pred = (model.predict(X_test) > 0.5).astype("int32")
    precision, recall, f1_score, _ = precision_recall_fscore_support(y_test,␣
↪y_pred)
    results.append((model_type, num_layers, dropout_rate, accuracy, precision,␣
↪recall, f1_score))
    print(f"Model Type: {model_type}, Num Layers: {num_layers}, Dropout Rate:␣
↪{dropout_rate}")
    print(f"Accuracy: {accuracy}, Precision: {precision}, Recall: {recall}, F1␣
↪Score: {f1_score}")
    evaluate_model(model, X_test, y_test)
```

```python
    print("------------------------------------------------------------------")

# Converting results to DataFrame
columns = ['Model Type', 'Num Layers', 'Dropout Rate', 'Accuracy', 'Precision',␣
 ↪'Recall', 'F1 Score']
results_df = pd.DataFrame(results, columns=columns)
print(results_df)
```

```
8/8 [==============================] - 1s 19ms/step
Model Type: RNN, Num Layers: 2, Dropout Rate: 0.3
Accuracy: 0.40400001406669617, Precision: [0.57309942 0.03797468 0.        ],
Recall: [0.80327869 0.5        0.        ], F1 Score: [0.66894198 0.07058824 0.
]
4/8 [==============>…] - ETA: 0s
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

```
8/8 [==============================] - 0s 20ms/step
              precision    recall  f1-score   support

           0       0.57      0.80      0.67       122
           1       0.04      0.50      0.07         6
           2       0.00      0.00      0.00       122

    accuracy                           0.40       250
   macro avg       0.20      0.43      0.25       250
weighted avg       0.28      0.40      0.33       250

```

```
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
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```

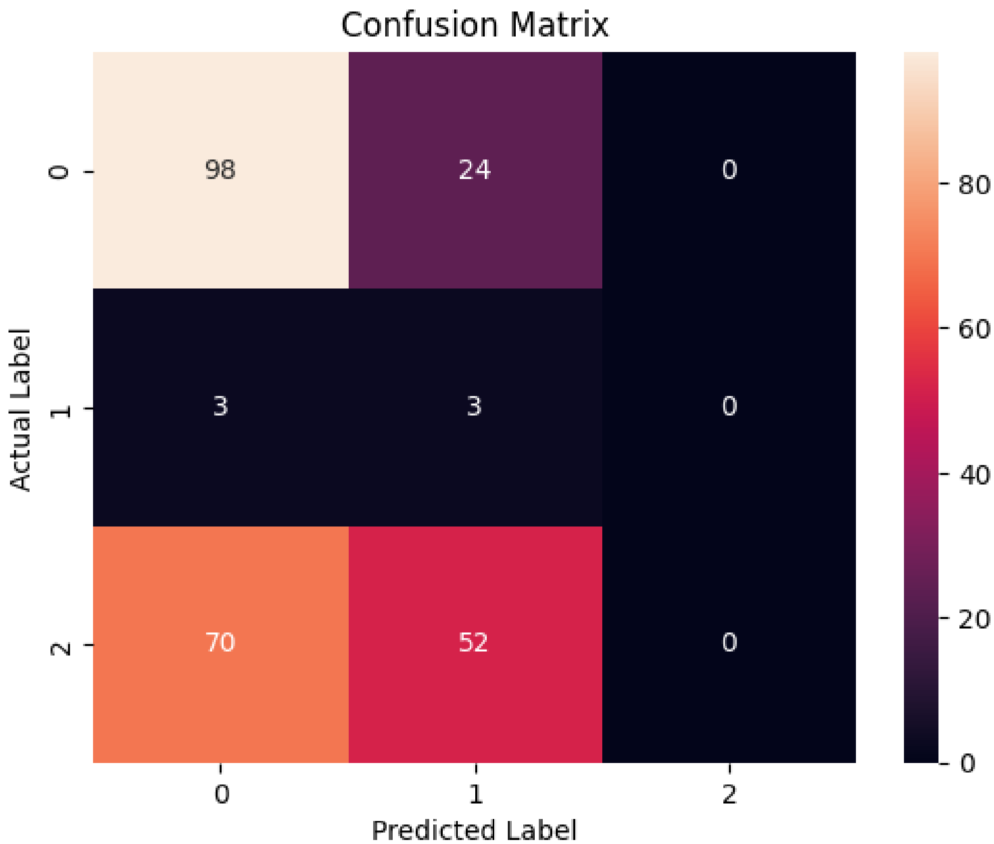


```
------------------------------------------------------------------
8/8 [==============================] - 2s 28ms/step
Model Type: RNN, Num Layers: 2, Dropout Rate: 0.7
Accuracy: 0.3880000114440918, Precision: [0.58641975 0.02272727 0.        ],
Recall: [0.77868852 0.33333333 0.        ], F1 Score: [0.66901408 0.04255319 0.
]
3/8 [==========>…] - ETA: 0s
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
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control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

```
8/8 [==============================] - 0s 29ms/step
              precision    recall  f1-score   support

           0       0.59      0.78      0.67       122
           1       0.02      0.33      0.04         6
           2       0.00      0.00      0.00       122
```

```
    accuracy                        0.39       250
   macro avg       0.20      0.37      0.24       250
weighted avg       0.29      0.39      0.33       250

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
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  _warn_prf(average, modifier, msg_start, len(result))
```

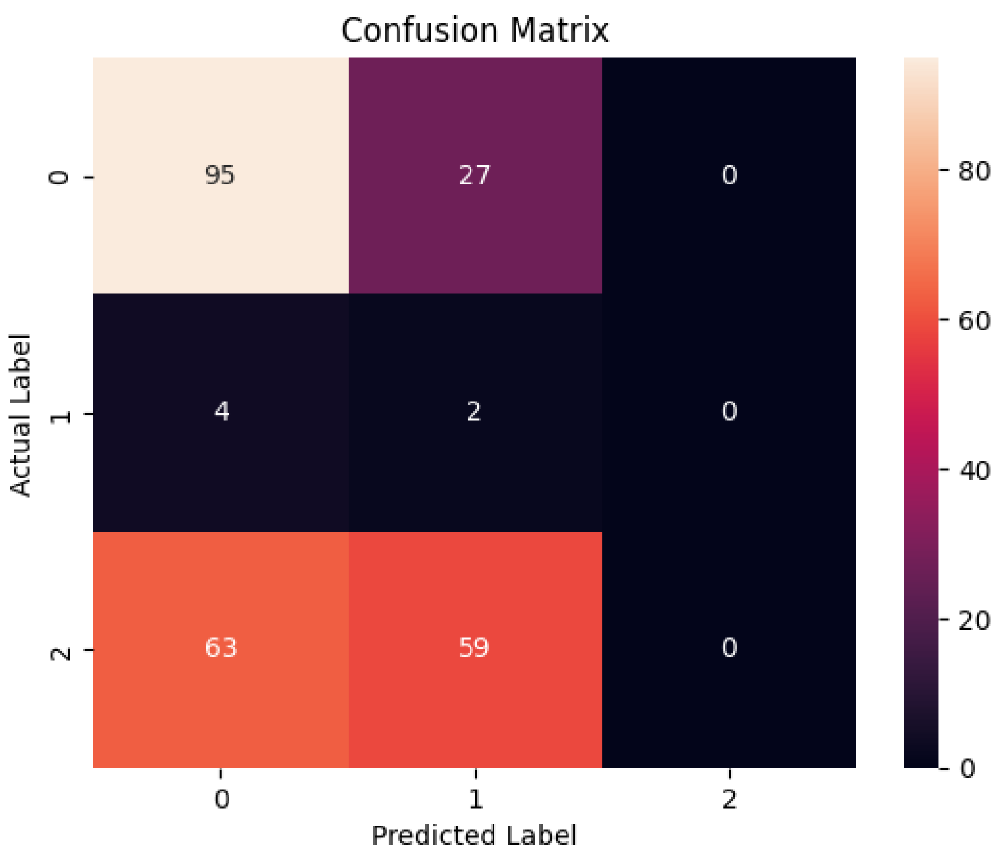

```
------------------------------------------------------------------
8/8 [==============================] - 1s 18ms/step
Model Type: GRU, Num Layers: 2, Dropout Rate: 0.3
Accuracy: 0.3479999899864197, Precision: [0.55333333 0.04       0.        ],
Recall: [0.68032787 0.66666667 0.        ], F1 Score: [0.61029412 0.0754717  0.
]
7/8 [========================>…] - ETA: 0s
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

```
8/8 [==============================] - 0s 18ms/step
              precision    recall  f1-score   support

           0       0.55      0.68      0.61       122
           1       0.04      0.67      0.08         6
           2       0.00      0.00      0.00       122

    accuracy                           0.35       250
   macro avg       0.20      0.45      0.23       250
weighted avg       0.27      0.35      0.30       250
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
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control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

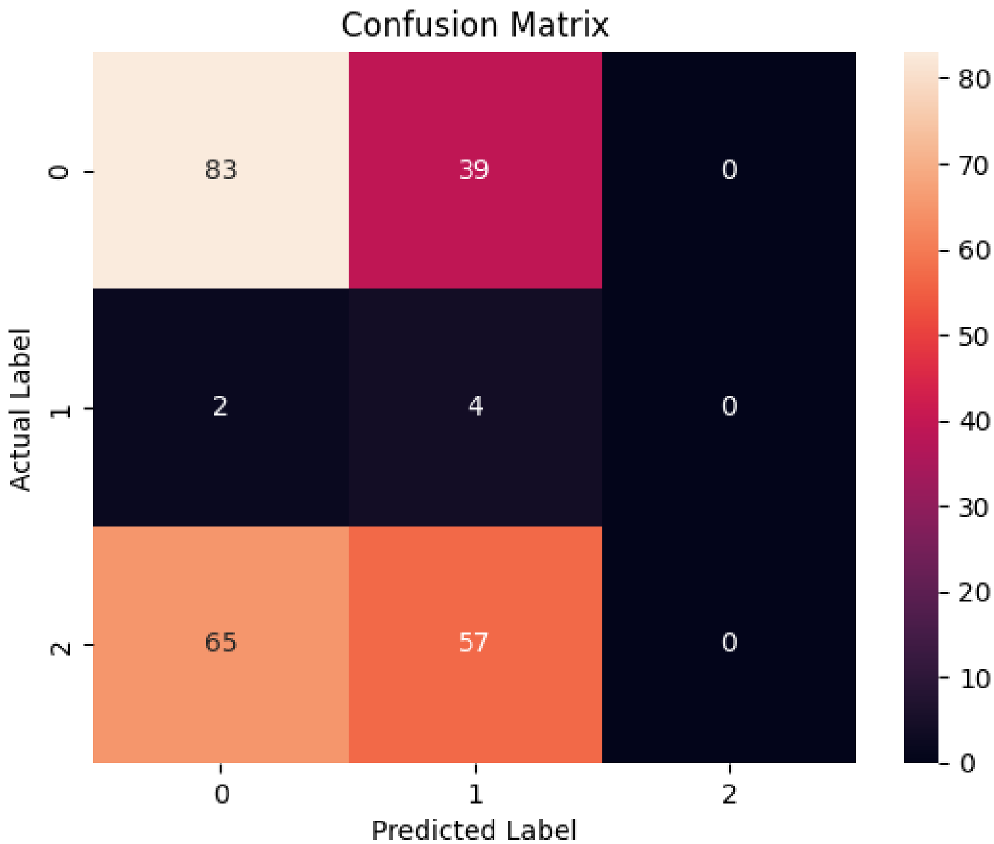


```
------------------------------------------------------------------
8/8 [==============================] - 1s 17ms/step
Model Type: GRU, Num Layers: 2, Dropout Rate: 0.7
Accuracy: 0.36000001430511475, Precision: [0.59722222 0.03773585 0.        ],
Recall: [0.70491803 0.66666667 0.        ], F1 Score: [0.64661654 0.07142857 0.
]
7/8 [=========================>…] - ETA: 0s

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))

8/8 [==============================] - 0s 18ms/step
              precision    recall  f1-score   support

           0       0.60      0.70      0.65       122
           1       0.04      0.67      0.07         6
           2       0.00      0.00      0.00       122
```

```
    accuracy                           0.36       250
   macro avg       0.21      0.46      0.24       250
weighted avg       0.29      0.36      0.32       250

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
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control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

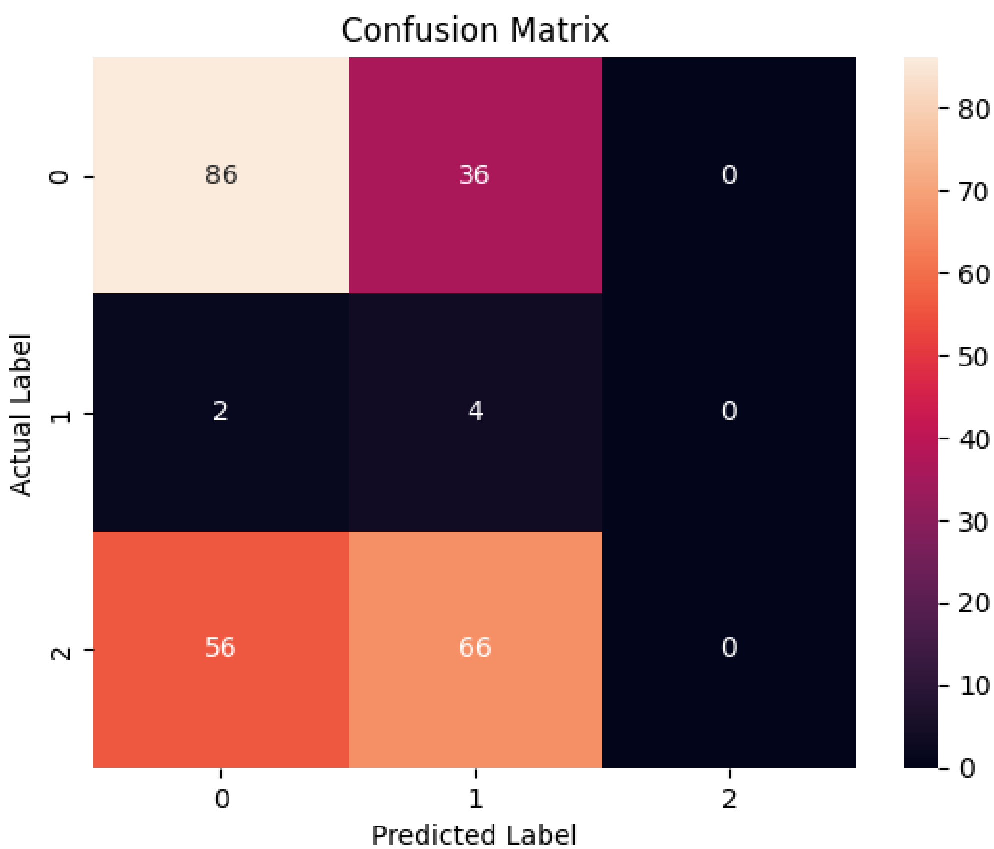

```
--------------------------------------------------------------------
8/8 [==============================] - 1s 20ms/step
Model Type: LSTM, Num Layers: 2, Dropout Rate: 0.3
Accuracy: 0.3199999928474426, Precision: [0.6015625  0.02459016 0.        ],
Recall: [0.63114754 0.5        0.        ], F1 Score: [0.616    0.046875 0.
]
4/8 [==============>…] - ETA: 0s
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

```
8/8 [==============================] - 0s 20ms/step
              precision    recall  f1-score   support

           0       0.60      0.63      0.62       122
           1       0.02      0.50      0.05         6
           2       0.00      0.00      0.00       122

    accuracy                           0.32       250
   macro avg       0.21      0.38      0.22       250
weighted avg       0.29      0.32      0.30       250
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
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  _warn_prf(average, modifier, msg_start, len(result))
```

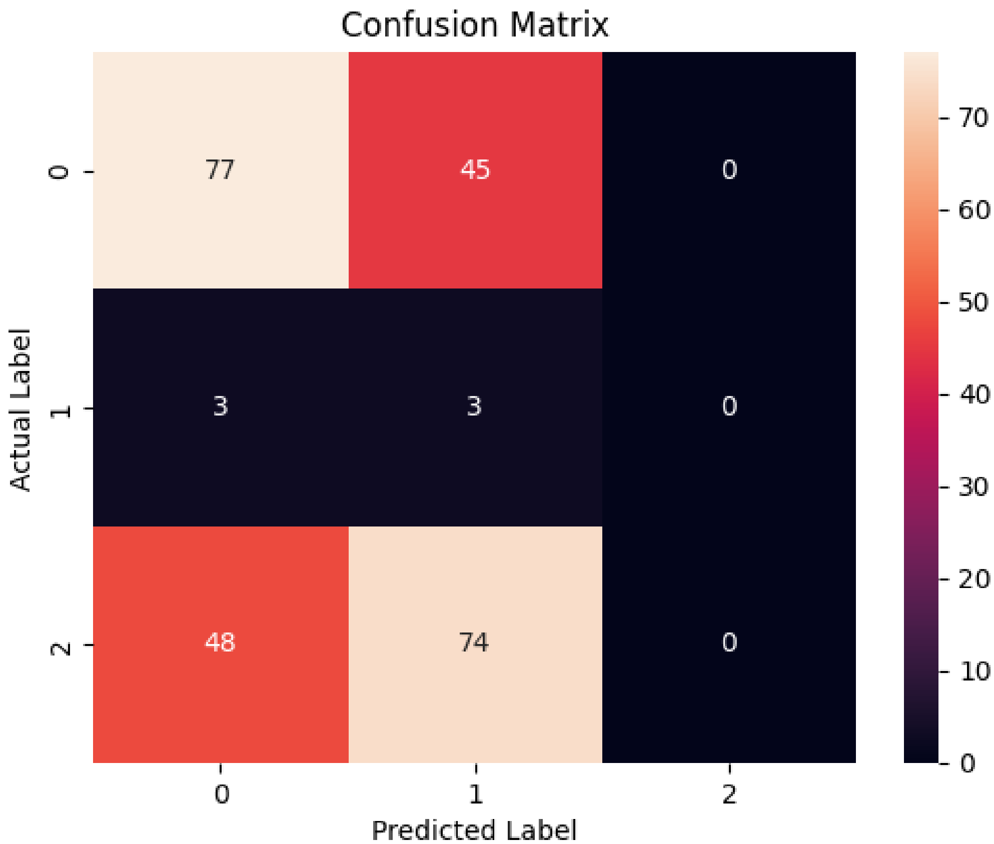


```
----------------------------------------------------------------
8/8 [==============================] - 2s 19ms/step
Model Type: LSTM, Num Layers: 2, Dropout Rate: 0.7
Accuracy: 0.19200000166893005, Precision: [0.66153846 0.02702703 0.        ],
Recall: [0.35245902 0.83333333 0.        ], F1 Score: [0.45989305 0.05235602 0.
]
4/8 [==============>…] - ETA: 0s

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))

8/8 [==============================] - 0s 20ms/step
              precision    recall  f1-score   support

           0       0.66      0.35      0.46       122
           1       0.03      0.83      0.05         6
           2       0.00      0.00      0.00       122
```

```
    accuracy                           0.19       250
   macro avg       0.23      0.40      0.17       250
weighted avg       0.32      0.19      0.23       250

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
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0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

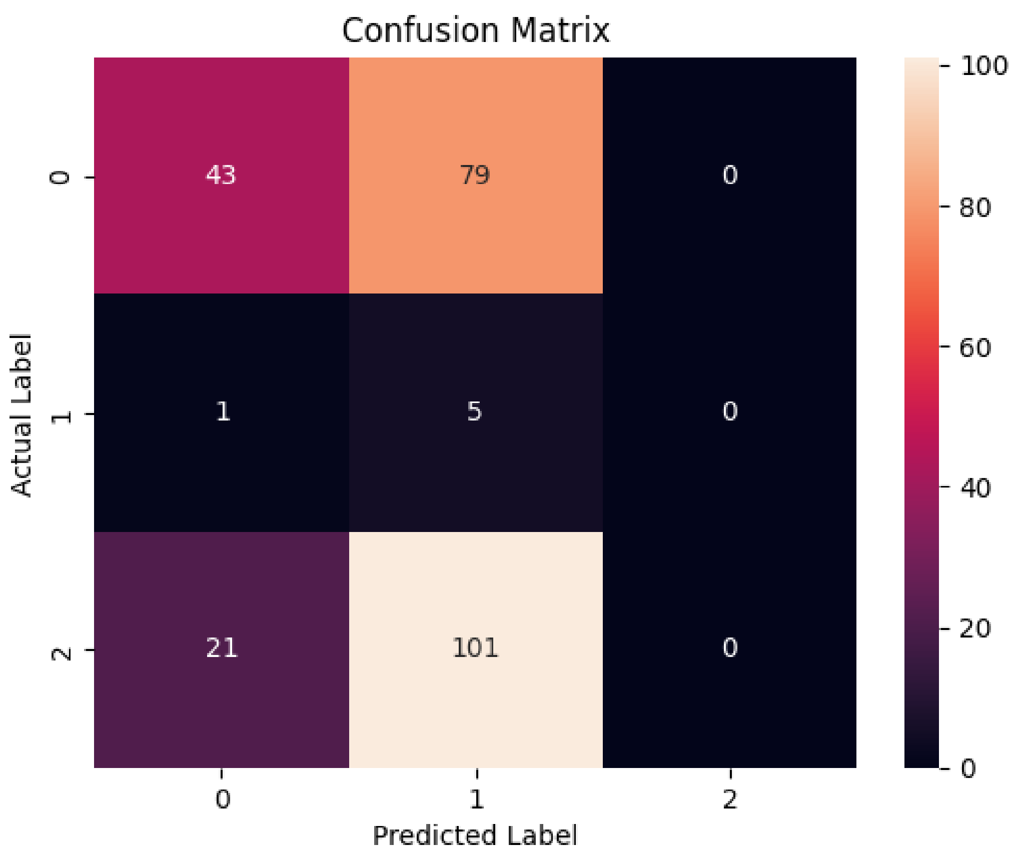

```
------------------------------------------------------------------
8/8 [==============================] - 3s 57ms/step
Model Type: BiLSTM, Num Layers: 2, Dropout Rate: 0.3
Accuracy: 0.20000000298023224, Precision: [0.64788732 0.02234637 0.        ],
Recall: [0.37704918 0.66666667 0.        ], F1 Score: [0.47668394 0.04324324 0.
]
1/8 [==>…] - ETA: 0s

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))

8/8 [==============================] - 0s 39ms/step

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
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/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))

              precision    recall  f1-score   support

           0       0.65      0.38      0.48       122
           1       0.02      0.67      0.04         6
           2       0.00      0.00      0.00       122

    accuracy                           0.20       250
   macro avg       0.22      0.35      0.17       250
weighted avg       0.32      0.20      0.23       250
```

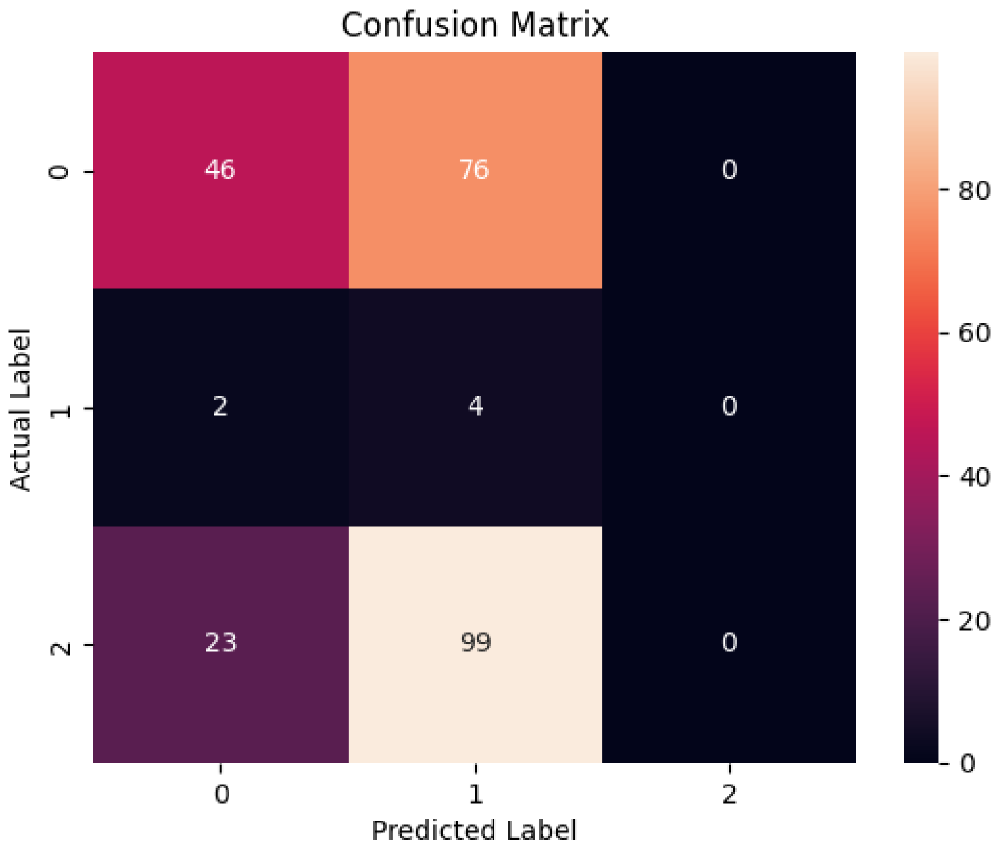


```
-----------------------------------------------------------------
8/8 [==============================] - 2s 32ms/step
Model Type: BiLSTM, Num Layers: 2, Dropout Rate: 0.7
Accuracy: 0.23999999463558197, Precision: [0.61290323 0.01910828 0.        ],
Recall: [0.46721311 0.5        0.        ], F1 Score: [0.53023256 0.03680982 0.
]
3/8 [==========>…] - ETA: 0s

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))

8/8 [==============================] - 0s 31ms/step
              precision    recall  f1-score   support

           0       0.61      0.47      0.53       122
           1       0.02      0.50      0.04         6
           2       0.00      0.00      0.00       122
```

```
    accuracy                           0.24       250
   macro avg       0.21      0.32      0.19       250
weighted avg       0.30      0.24      0.26       250

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
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0.0 in labels with no predicted samples. Use `zero_division` parameter to
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```

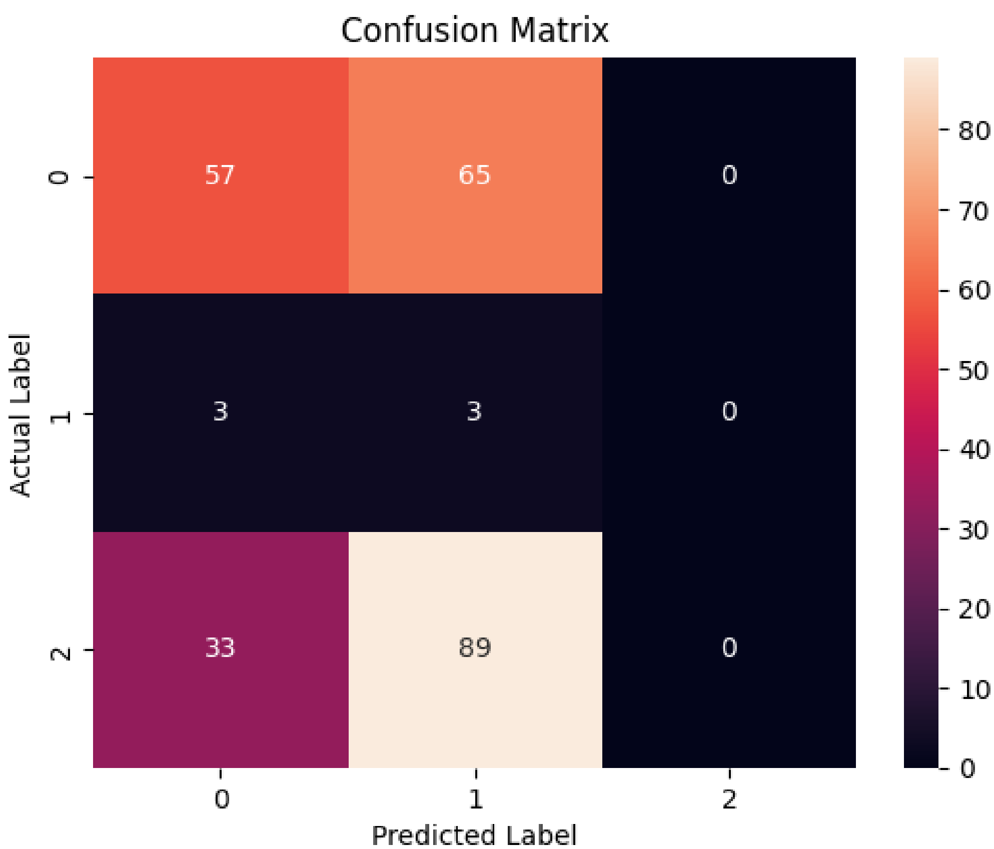

```
------------------------------------------------------------------
8/8 [==============================] - 2s 24ms/step
Model Type: RNN, Num Layers: 3, Dropout Rate: 0.3
Accuracy: 0.36000001430511475, Precision: [0.5398773  0.02298851 0.        ],
Recall: [0.72131148 0.33333333 0.        ], F1 Score: [0.61754386 0.04301075 0.
]
4/8 [==============>…] - ETA: 0s

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))

8/8 [==============================] - 0s 24ms/step
              precision    recall  f1-score   support

           0       0.54      0.72      0.62       122
           1       0.02      0.33      0.04         6
           2       0.00      0.00      0.00       122

    accuracy                           0.36       250
   macro avg       0.19      0.35      0.22       250
weighted avg       0.26      0.36      0.30       250


/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
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/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

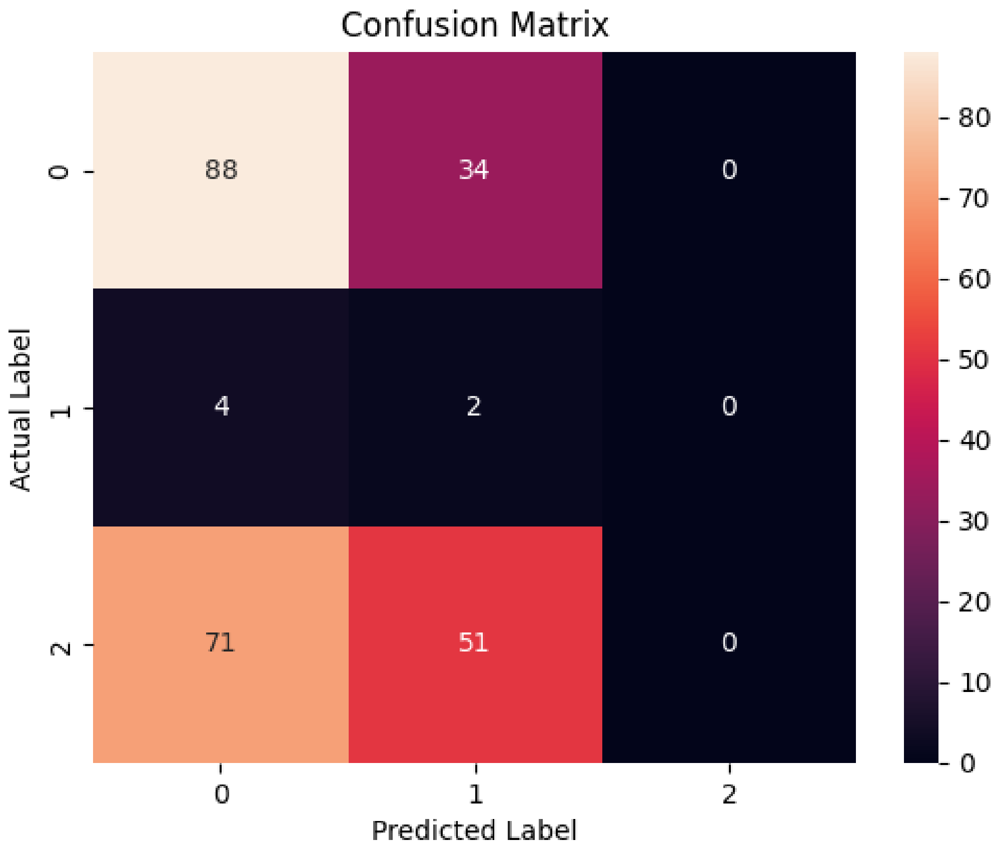


```
----------------------------------------------------------------------
8/8 [==============================] - 4s 36ms/step
Model Type: RNN, Num Layers: 3, Dropout Rate: 0.7
Accuracy: 0.3160000145435333, Precision: [0.57777778 0.00869565 0.         ],
Recall: [0.63934426 0.16666667 0.         ], F1 Score: [0.60700389 0.01652893 0.
]
1/8 [==>…] - ETA: 0s

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))

8/8 [==============================] - 0s 25ms/step
              precision    recall  f1-score   support

           0       0.58      0.64      0.61       122
           1       0.01      0.17      0.02         6
           2       0.00      0.00      0.00       122
```

```
    accuracy                           0.32       250
   macro avg       0.20      0.27      0.21       250
weighted avg       0.28      0.32      0.30       250

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
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/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

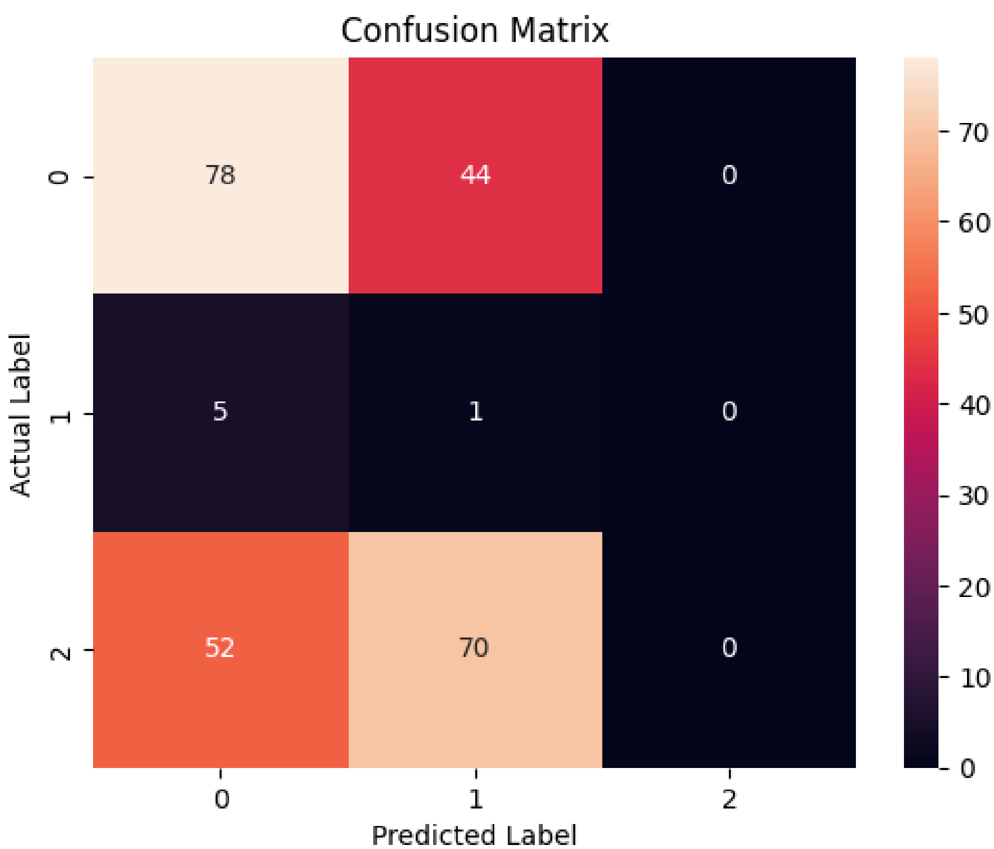

```
--------------------------------------------------------------------
8/8 [==============================] - 2s 22ms/step
Model Type: GRU, Num Layers: 3, Dropout Rate: 0.3
Accuracy: 0.40400001406669617, Precision: [0.54748603 0.04225352 0.        ],
Recall: [0.80327869 0.5        0.        ], F1 Score: [0.65116279 0.07792208 0.
]
4/8 [==============>…] - ETA: 0s

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))

8/8 [==============================] - 0s 22ms/step
              precision    recall  f1-score   support

           0       0.55      0.80      0.65       122
           1       0.04      0.50      0.08         6
           2       0.00      0.00      0.00       122

    accuracy                           0.40       250
   macro avg       0.20      0.43      0.24       250
weighted avg       0.27      0.40      0.32       250


/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
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  _warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

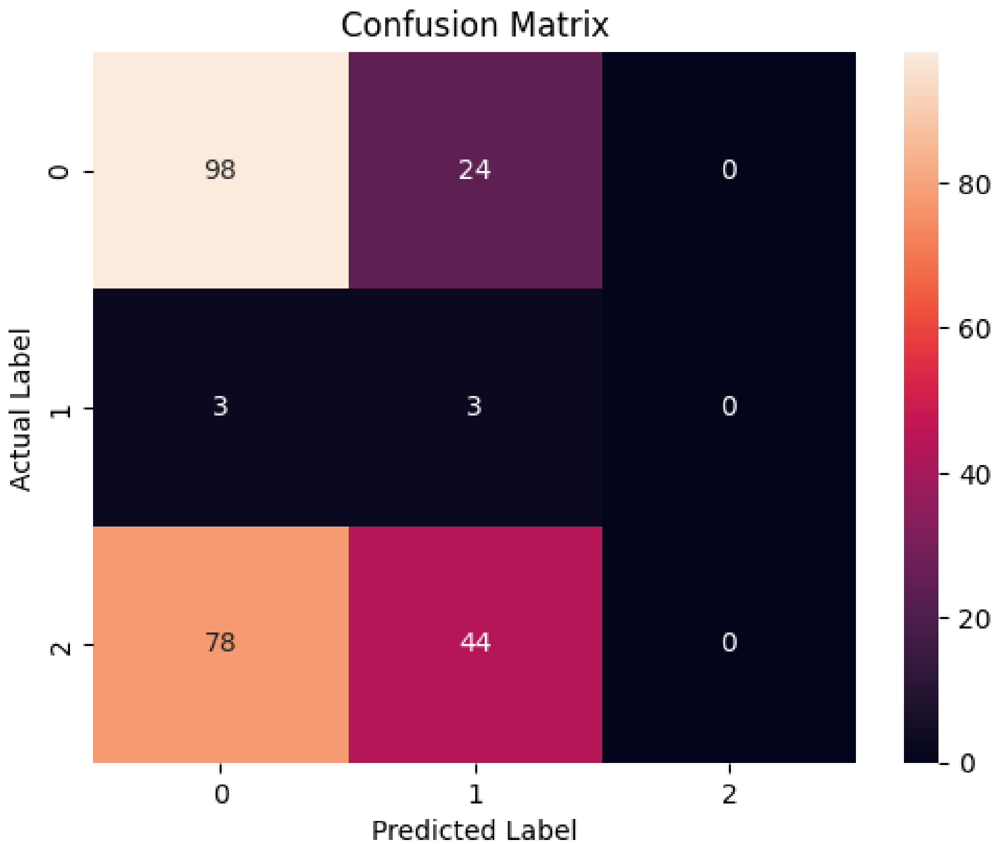


```
----------------------------------------------------------------
8/8 [==============================] - 2s 22ms/step
Model Type: GRU, Num Layers: 3, Dropout Rate: 0.7
Accuracy: 0.3880000114440918, Precision: [0.51612903 0.015625   0.        ],
Recall: [0.78688525 0.16666667 0.        ], F1 Score: [0.62337662 0.02857143 0.
]
3/8 [==========>…] - ETA: 0s
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

```
8/8 [==============================] - 0s 24ms/step
              precision    recall  f1-score   support

           0       0.52      0.79      0.62       122
           1       0.02      0.17      0.03         6
           2       0.00      0.00      0.00       122
```

```
    accuracy                           0.39       250
   macro avg       0.18      0.32      0.22       250
weighted avg       0.25      0.39      0.30       250


/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
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/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```
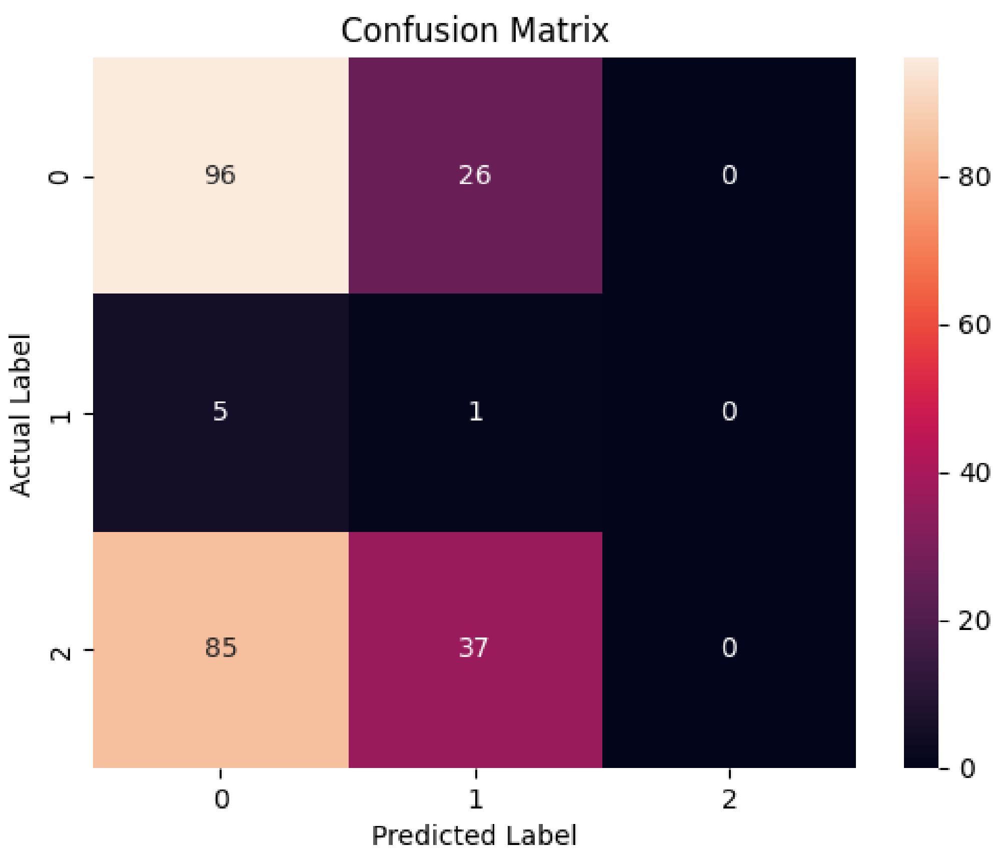
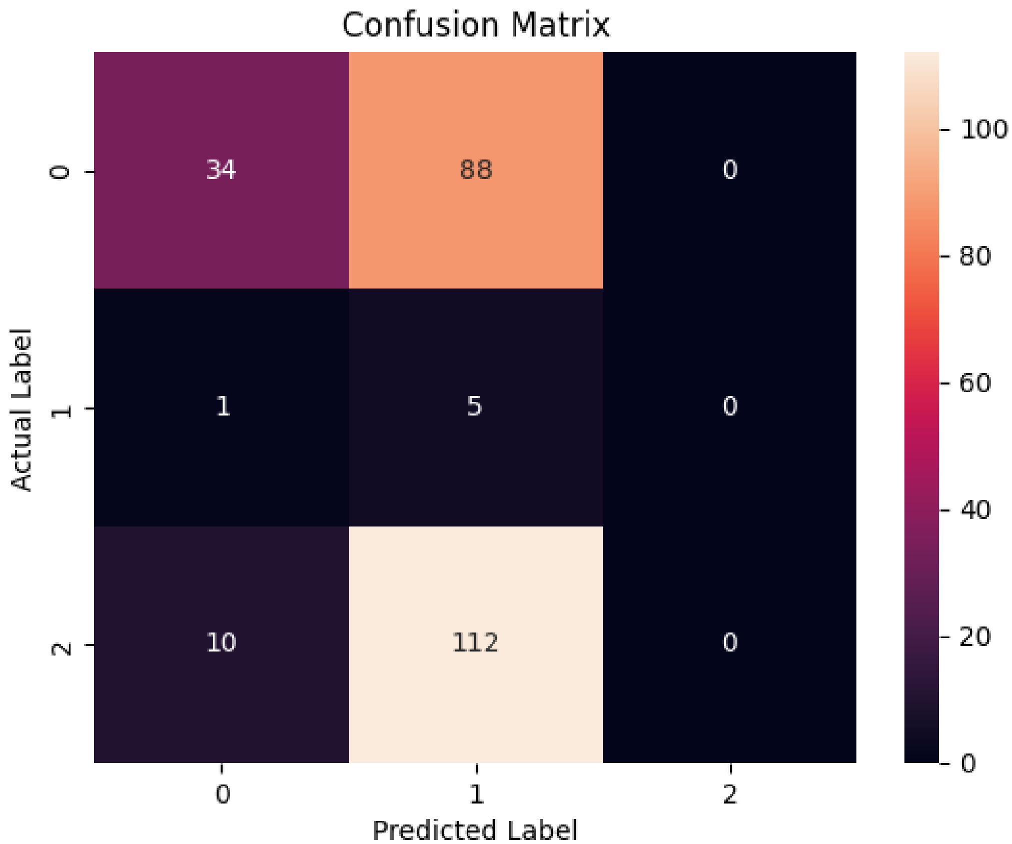



```
-------------------------------------------------------------------
8/8 [==============================] - 2s 24ms/step
Model Type: LSTM, Num Layers: 3, Dropout Rate: 0.3
Accuracy: 0.15600000321865082, Precision: [0.75555556 0.02439024 0.        ],
Recall: [0.27868852 0.83333333 0.        ], F1 Score: [0.40718563 0.04739336 0.
]
5/8 [=================>…] - ETA: 0s
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

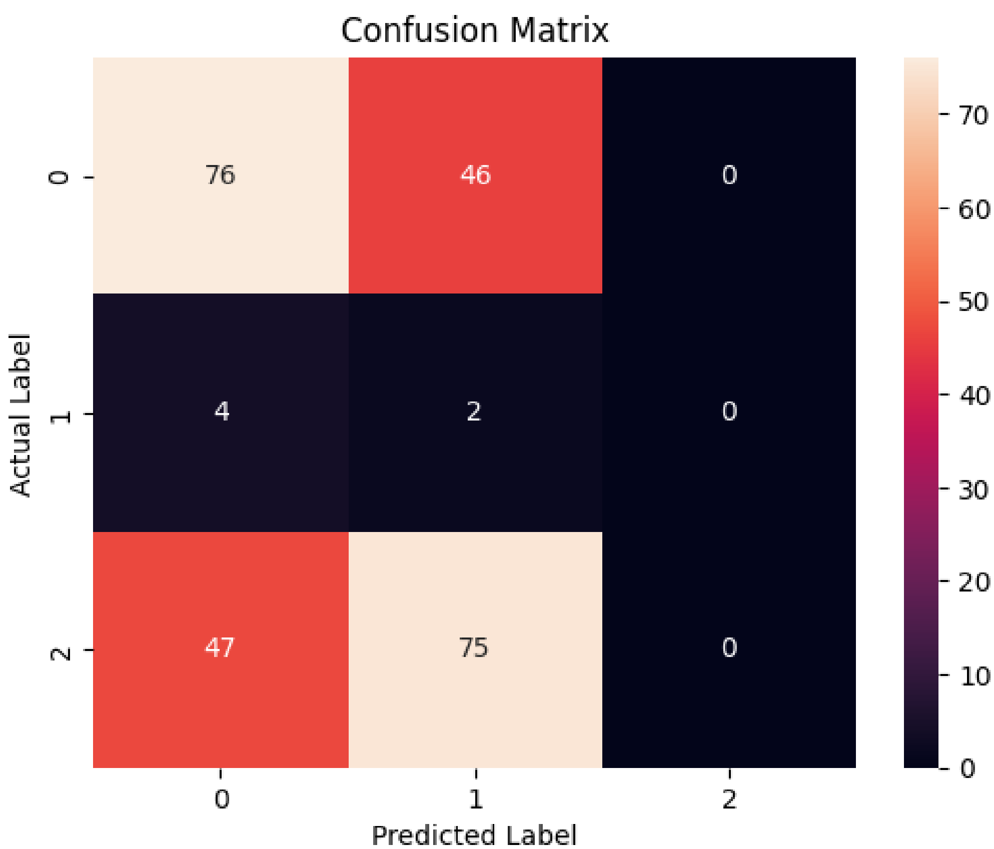
```
8/8 [==============================] - 0s 26ms/step
              precision    recall  f1-score   support

           0       0.76      0.28      0.41       122
           1       0.02      0.83      0.05         6
           2       0.00      0.00      0.00       122

    accuracy                           0.16       250
   macro avg       0.26      0.37      0.15       250
weighted avg       0.37      0.16      0.20       250
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

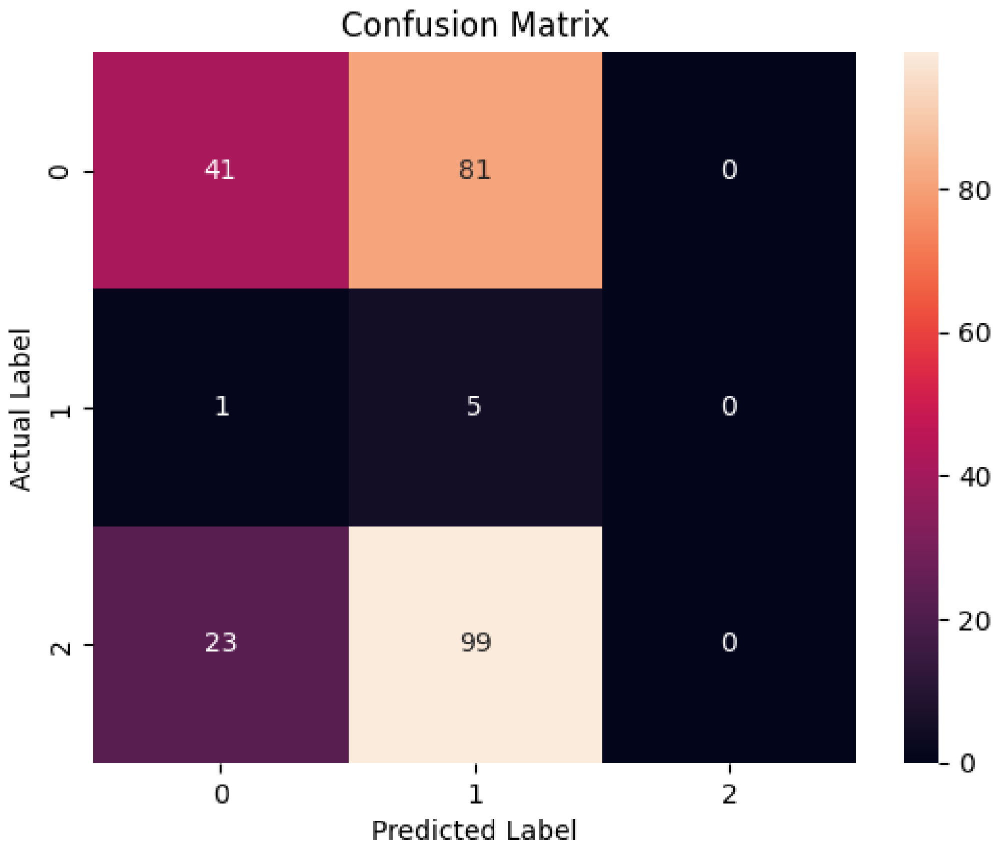



```
------------------------------------------------------------------
8/8 [==============================] - 2s 27ms/step
Model Type: LSTM, Num Layers: 3, Dropout Rate: 0.7
Accuracy: 0.31200000643730164, Precision: [0.5984252  0.01626016 0.        ],
Recall: [0.62295082 0.33333333 0.        ], F1 Score: [0.61044177 0.03100775 0.
]
3/8 [==========>…] - ETA: 0s

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))

8/8 [==============================] - 0s 29ms/step
              precision    recall  f1-score   support

           0       0.60      0.62      0.61       122
           1       0.02      0.33      0.03         6
           2       0.00      0.00      0.00       122
```

```
    accuracy                           0.31       250
   macro avg       0.20      0.32      0.21       250
weighted avg       0.29      0.31      0.30       250

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

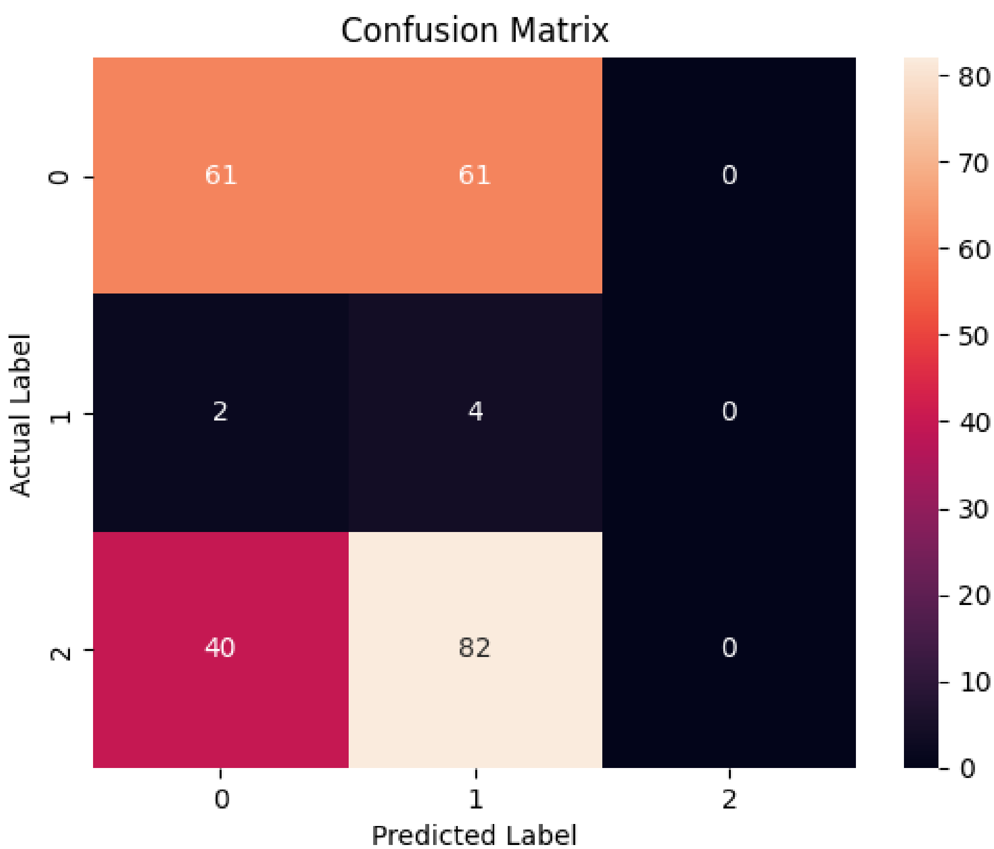


```
------------------------------------------------------------------
8/8 [==============================] - 3s 41ms/step
Model Type: BiLSTM, Num Layers: 3, Dropout Rate: 0.3
Accuracy: 0.18400000035762787, Precision: [0.63076923 0.02702703 0.        ],
Recall: [0.33606557 0.83333333 0.        ], F1 Score: [0.43850267 0.05235602 0.
]
2/8 [======>…] - ETA: 0s

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))

8/8 [==============================] - 0s 46ms/step
              precision    recall  f1-score   support

           0       0.63      0.34      0.44       122
           1       0.03      0.83      0.05         6
           2       0.00      0.00      0.00       122

    accuracy                           0.18       250
   macro avg       0.22      0.39      0.16       250
weighted avg       0.31      0.18      0.22       250


/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```




```
-------------------------------------------------------------------
8/8 [==============================] - 3s 56ms/step
Model Type: BiLSTM, Num Layers: 3, Dropout Rate: 0.7
Accuracy: 0.25999999046325684, Precision: [0.59223301 0.02721088 0.        ],
Recall: [0.5        0.66666667 0.        ], F1 Score: [0.54222222 0.05228758 0.
]
1/8 [==>…] - ETA: 0s

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))

8/8 [==============================] - 1s 73ms/step
              precision    recall  f1-score   support

           0       0.59      0.50      0.54       122
           1       0.03      0.67      0.05         6
           2       0.00      0.00      0.00       122
```

```
    accuracy                           0.26       250
   macro avg       0.21      0.39      0.20       250
weighted avg       0.29      0.26      0.27       250

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision and F-score are ill-defined and being set to
0.0 in labels with no predicted samples. Use `zero_division` parameter to
control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```



```
------------------------------------------------------------------
   Model Type  Num Layers  Dropout Rate  Accuracy  \
0         RNN           2           0.3     0.404
1         RNN           2           0.7     0.388
2         GRU           2           0.3     0.348
3         GRU           2           0.7     0.360
4        LSTM           2           0.3     0.320
5        LSTM           2           0.7     0.192
6      BiLSTM           2           0.3     0.200
7      BiLSTM           2           0.7     0.240
8         RNN           3           0.3     0.360
9         RNN           3           0.7     0.316
10        GRU           3           0.3     0.404
11        GRU           3           0.7     0.388
12       LSTM           3           0.3     0.156
13       LSTM           3           0.7     0.312
14     BiLSTM           3           0.3     0.184
15     BiLSTM           3           0.7     0.260

                                           Precision  \
0     [0.5730994152046783, 0.0379746835443038, 0.0]
1   [0.5864197530864198, 0.022727272727272728, 0.0]
2                   [0.5533333333333333, 0.04, 0.0]
3    [0.5972222222222222, 0.03773584905660377, 0.0]
4              [0.6015625, 0.02459016393442623, 0.0]
5    [0.6615384615384615, 0.02702702702702703, 0.0]
6      [0.647887323943662, 0.0223463687150838, 0.0]
7    [0.6129032258064516, 0.01910828025477707, 0.0]
8   [0.5398773006134969, 0.022988505747126436, 0.0]
9   [0.5777777777777777, 0.008695652173913044, 0.0]
10    [0.547486033519553, 0.04225352112676056, 0.0]
11               [0.5161290322580645, 0.015625, 0.0]
12  [0.7555555555555555, 0.024390243902439025, 0.0]
13  [0.5984251968503937, 0.016260162601626018, 0.0]
14   [0.6307692307692307, 0.02702702702702703, 0.0]
15  [0.5922330097087378, 0.027210884353741496, 0.0]

                                             Recall  \
0                   [0.8032786885245902, 0.5, 0.0]
1    [0.7786885245901639, 0.3333333333333333, 0.0]
2     [0.680327868852459, 0.6666666666666666, 0.0]
3    [0.7049180327868853, 0.6666666666666666, 0.0]
4                   [0.6311475409836066, 0.5, 0.0]
5    [0.3524590163934426, 0.8333333333333334, 0.0]
6    [0.3770491803278688, 0.6666666666666666, 0.0]
7                   [0.4672131147540984, 0.5, 0.0]
8    [0.7213114754098361, 0.3333333333333333, 0.0]
9    [0.639344262295082, 0.16666666666666666, 0.0]
```

```
10                  [0.8032786885245902, 0.5, 0.0]
11   [0.7868852459016393, 0.16666666666666666, 0.0]
12    [0.2786885245901639, 0.8333333333333334, 0.0]
13    [0.6229508196721312, 0.3333333333333333, 0.0]
14    [0.3360655737704918, 0.8333333333333334, 0.0]
15                  [0.5, 0.6666666666666666, 0.0]

                                               F1 Score
0      [0.6689419795221841, 0.07058823529411765, 0.0]
1      [0.6690140845070423, 0.04255319148936171, 0.0]
2      [0.6102941176470589, 0.07547169811320754, 0.0]
3      [0.6466165413533835, 0.07142857142857142, 0.0]
4                            [0.616, 0.046875, 0.0]
5     [0.45989304812834225, 0.05235602094240838, 0.0]
6    [0.47668393782383417, 0.043243243243243246, 0.0]
7      [0.5302325581395348, 0.03680981595092024, 0.0]
8      [0.6175438596491228, 0.04301075268817204, 0.0]
9      [0.6070038910505836, 0.01652892561983471, 0.0]
10     [0.6511627906976744, 0.07792207792207793, 0.0]
11     [0.6233766233766234, 0.02857142857142857, 0.0]
12    [0.4071856287425149, 0.047393364928909956, 0.0]
13      [0.6104417670682731, 0.0310077519379845, 0.0]
14     [0.4385026737967914, 0.05235602094240838, 0.0]
15      [0.5422222222222222, 0.0522875816993464, 0.0]
```

[54]:
```python
print("Model Evaluation Results:")
print(results_df.to_string(index=False))

# Finding the best performing model
best_model_idx = results_df['Accuracy'].idxmax()
best_model_row = results_df.iloc[best_model_idx]
best_model_params = best_model_row[['Model Type', 'Num Layers', 'Dropout Rate']]
best_model_accuracy = best_model_row['Accuracy']

print("\nBest Performing Model:")
print(f"Model Type: {best_model_params['Model Type']}")
print(f"Number of Layers: {best_model_params['Num Layers']}")
print(f"Dropout Rate: {best_model_params['Dropout Rate']}")
print(f"Accuracy: {best_model_accuracy}")
```

```
Model Evaluation Results:
Model Type  Num Layers  Dropout Rate  Accuracy
Precision                                           Recall
F1 Score
       RNN           2           0.3     0.404   [0.5730994152046783,
0.0379746835443038, 0.0]                 [0.8032786885245902, 0.5, 0.0]
[0.6689419795221841, 0.07058823529411765, 0.0]
       RNN           2           0.7     0.388  [0.5864197530864198,
```

```
0.022727272727272728, 0.0]  [0.7786885245901639, 0.3333333333333333, 0.0]
[0.6690140845070423, 0.04255319148936171, 0.0]
       GRU           2           0.3     0.348
[0.5533333333333333, 0.04, 0.0]   [0.680327868852459, 0.6666666666666666, 0.0]
[0.6102941176470589, 0.07547169811320754, 0.0]
       GRU           2           0.7     0.360  [0.5972222222222222,
0.03773584905660377, 0.0]  [0.7049180327868853, 0.6666666666666666, 0.0]
[0.6466165413533835, 0.07142857142857142, 0.0]
      LSTM           2           0.3     0.320           [0.6015625,
0.02459016393442623, 0.0]                 [0.6311475409836066, 0.5, 0.0]
[0.616, 0.046875, 0.0]
      LSTM           2           0.7     0.192  [0.6615384615384615,
0.02702702702702703, 0.0]  [0.3524590163934426, 0.8333333333333334, 0.0]
[0.45989304812834225, 0.05235602094240838, 0.0]
    BiLSTM           2           0.3     0.200    [0.647887323943662,
0.0223463687150838, 0.0]  [0.3770491803278688, 0.6666666666666666, 0.0]
[0.47668393782383417, 0.043243243243243246, 0.0]
    BiLSTM           2           0.7     0.240  [0.6129032258064516,
0.01910828025477707, 0.0]                 [0.4672131147540984, 0.5, 0.0]
[0.5302325581395348, 0.03680981595092024, 0.0]
       RNN           3           0.3     0.360 [0.5398773006134969,
0.022988505747126436, 0.0]  [0.7213114754098361, 0.3333333333333333, 0.0]
[0.6175438596491228, 0.04301075268817204, 0.0]
       RNN           3           0.7     0.316 [0.5777777777777777,
0.008695652173913044, 0.0]  [0.639344262295082, 0.16666666666666666, 0.0]
[0.6070038910505836, 0.01652892561983471, 0.0]
       GRU           3           0.3     0.404   [0.547486033519553,
0.04225352112676056, 0.0]                 [0.8032786885245902, 0.5, 0.0]
[0.6511627906976744, 0.07792207792207793, 0.0]
       GRU           3           0.7     0.388              [0.5161290322580645,
0.015625, 0.0] [0.7868852459016393, 0.16666666666666666, 0.0]
[0.6233766233766234, 0.02857142857142857, 0.0]
      LSTM           3           0.3     0.156 [0.7555555555555555,
0.024390243902439025, 0.0]  [0.2786885245901639, 0.8333333333333334, 0.0]
[0.4071856287425149, 0.047393364928909956, 0.0]
      LSTM           3           0.7     0.312 [0.5984251968503937,
0.016260162601626018, 0.0]  [0.6229508196721312, 0.3333333333333333, 0.0]
[0.6104417670682731, 0.0310077519379845, 0.0]
    BiLSTM           3           0.3     0.184  [0.6307692307692307,
0.02702702702702703, 0.0]  [0.3360655737704918, 0.8333333333333334, 0.0]
[0.4385026737967914, 0.05235602094240838, 0.0]
    BiLSTM           3           0.7     0.260 [0.5922330097087378,
0.027210884353741496, 0.0]                 [0.5, 0.6666666666666666, 0.0]
[0.5422222222222222, 0.0522875816993464, 0.0]

Best Performing Model:
Model Type: RNN
Number of Layers: 2
```

```
Dropout Rate: 0.3
Accuracy: 0.40400001406669617
```

4 Question 2

Note: The dataset used in this analysis has been preprocessed in Question 1, and the preprocessed version is utilized here for training and testing.

[55]:
```
# Store the best performing model
best_model = model

from sklearn.metrics import precision_score, recall_score

# Function to build and train a classifier with specified embedding
def build_and_train_embedding_classifier(embedding_type, X_train, y_train,␣
 ↪X_test, y_test):
    # Build the model
    embedding_dim = 100  # Adjust dimension based on the chosen embedding method
    embedding_model = build_model(best_model_type, best_num_layers,␣
 ↪best_dropout_rate)

    # Train the model
    history = embedding_model.fit(X_train, y_train, epochs=8, batch_size=32,␣
 ↪validation_split=0.25, verbose=0)

    # Evaluate the model
    y_pred = (embedding_model.predict(X_test) > 0.5).astype("int32")
    accuracy = accuracy_score(y_test, y_pred)
    precision = precision_score(y_test, y_pred, average='weighted')
    recall = recall_score(y_test, y_pred, average='weighted')
    f1_value = calculate_f1_score(y_test, y_pred, average='weighted')

    return accuracy, precision, recall, f1_value

# Perform binary classification with different embeddings
embedding_results = {}

# Example: using Word2Vec embedding
word2vec_accuracy, word2vec_precision, word2vec_recall, word2vec_f1_score =␣
 ↪build_and_train_embedding_classifier('Word2Vec', X_train, y_train, X_test,␣
 ↪y_test)
embedding_results['Word2Vec'] = [word2vec_accuracy, word2vec_precision,␣
 ↪word2vec_recall, word2vec_f1_score]

# Similarly, perform classification with other embedding methods
# Example: using Glove embedding
```

```
glove_accuracy, glove_precision, glove_recall, glove_f1_score =␣
↪build_and_train_embedding_classifier('Glove', X_train, y_train, X_test,␣
↪y_test)
embedding_results['Glove'] = [glove_accuracy, glove_precision, glove_recall,␣
↪glove_f1_score]

# Example: using FastText embedding
fasttext_accuracy, fasttext_precision, fasttext_recall, fasttext_f1_score =␣
↪build_and_train_embedding_classifier('FastText', X_train, y_train, X_test,␣
↪y_test)
embedding_results['FastText'] = [fasttext_accuracy, fasttext_precision,␣
↪fasttext_recall, fasttext_f1_score]

# Example: using Elmo embedding
elmo_accuracy, elmo_precision, elmo_recall, elmo_f1_score =␣
↪build_and_train_embedding_classifier('Elmo', X_train, y_train, X_test,␣
↪y_test)
embedding_results['Elmo'] = [elmo_accuracy, elmo_precision, elmo_recall,␣
↪elmo_f1_score]

# Example: using model without embeddings
no_embedding_accuracy, no_embedding_precision, no_embedding_recall,␣
↪no_embedding_f1_score = build_and_train_embedding_classifier('', X_train,␣
↪y_train, X_test, y_test)
embedding_results['Without Embeddings'] = [no_embedding_accuracy,␣
↪no_embedding_precision, no_embedding_recall, no_embedding_f1_score]

# Print results
print("Results for different embeddings:")
embedding_df = pd.DataFrame(embedding_results, index=['Accuracy', 'Precision',␣
↪'Recall', 'F1 Score']).transpose()
print(embedding_df)
```

```
8/8 [==============================] - 1s 18ms/step

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, msg_start, len(result))

8/8 [==============================] - 1s 19ms/step

/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

```
8/8 [==============================] - 1s 16ms/step
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

```
8/8 [==============================] - 1s 19ms/step
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

```
8/8 [==============================] - 1s 15ms/step
Results for different embeddings:
                    Accuracy  Precision  Recall  F1 Score
Word2Vec               0.272   0.300965   0.272  0.277653
Glove                  0.172   0.346529   0.172  0.216006
FastText               0.424   0.282535   0.424  0.335380
Elmo                   0.296   0.300355   0.296  0.290844
Without Embeddings     0.308   0.297605   0.308  0.295825
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/metrics/_classification.py:1344:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```